

PHYS 350: Honours Electricity and Magnetism

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Abstract

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1 Introduction

Textbook used is David J. Griffiths - Introduction to Electrodynamics Fifth Edition. Course grading is divided as follows:

- 35% problem sets (must be handwritten scans).
- 25% midterm.
- 40% final exam.

2 Vector Analysis

2.1 Terminologies

Definition 1 (Scalar). *A scalar is a quantity with just a magnitude.*

Definition 2 (Vector). *A vector is a quantity with a magnitude and a direction, i.e. $\vec{v} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z}$.*

Definition 3 (Cross Product). *The cross product is a vector perpendicular to both, set by the right hand rule, and it vanishes for parallel vectors:*

$$\vec{v} \times \vec{w} = |\vec{v}||\vec{w}| \sin \theta \hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \hat{x}(v_y w_z - w_y v_z) - \hat{y}(v_x w_z - w_x v_z) + \hat{z}(v_x w_y - w_x v_y).$$

Definition 4 (Dot Product). *The dot product measures how much of $\vec{v}$ points along $\vec{w}$, so it is zero for perpendicular vectors:*

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z = |\vec{v}||\vec{w}| \cos \theta.$$

Remark 1. Note $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, and $|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}$.

Definition 5 (Field). *A field is something with a value at every point in space. A scalar field assigns a number to each point, like temperature $T(x, y, z)$ or the height of a hill. A vector field assigns a vector to each point, like fluid velocity or $\vec{E}$.*

2.2 Differential Operators

Definition 6 (Del operator). *The del operator*

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right).$$

Definition 7 (Gradient). *The gradient turns a scalar into a vector, pointing in the direction of steepest increase of f :*

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}.$$

Remark 2. The gradient points in the steepest uphill direction, and its length grows where the hill is steeper. It is always perpendicular to the contours of constant f .

Definition 8 (Divergence). Divergence turns a vector into a scalar, measuring sources and sinks. For a small box, non-zero divergence means the inflow and outflow through its faces do not balance: arrows spreading from a point (positive) mean fluid is created there, arrows converging (negative) mean it is destroyed.

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

Definition 9 (Curl). Curl turns a vector into a vector, measuring local rotation. Picture a small paddle wheel: if the flow spins it, the curl is non-zero there. Uniform flow does not spin it, but shear flow (parallel arrows, longer on one side) does, so straight field lines do not imply zero curl.

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}.$$

2.3 Vector Integrals

Definition 10 (Integral). An integral is an instruction to move along a region, evaluate something at each step, and add it to a running total.

Definition 11 (Line Integral). A line integral sums $\vec{F}$ along a curve, picking up only the component tangent to the path:

$$\int_C \vec{F} \cdot d\vec{l}.$$

Example. Take $\vec{F} = y\hat{x}$ along a counterclockwise quarter circle of radius R . On the circle $d\vec{l} = R d\phi(-\sin\phi\hat{x} + \cos\phi\hat{y})$ and $y = R\sin\phi$, so

$$\int_C \vec{F} \cdot d\vec{l} = -R^2 \int_0^{\pi/2} \sin^2\phi d\phi = -\frac{R^2\pi}{4}.$$

It is negative because $\vec{F}$ points along $+\hat{x}$ while we travel in the $-\hat{x}$ direction over most of the arc.

Definition 12 (Flux). Flux measures how much of the field pokes through a surface:

$$\int_S \vec{F} \cdot d\vec{a} = \int_S \vec{F} \cdot \hat{n} da.$$

Definition 13 (Volume Integral). A volume integral sums a scalar over a region, giving the mass if f is a density:

$$\int_V f d^3\vec{r} = \int_V f dx dy dz.$$

Remark 3. We write $\oint$ when the curve is a closed loop or the surface is closed.

2.4 The Three Fundamental Theorems

Integrating a derivative over a region reduces to evaluating the original object on the boundary of that region.

Theorem 1 (Gradient theorem). *The boundary of a curve is two points:*

$$\int_a^b \vec{\nabla} h \cdot d\vec{l} = h(b) - h(a).$$

To find my altitude change climbing a hill, I can compare before and after, or add up all the little changes along the way.

Theorem 2 (Stokes' theorem). *The boundary of a surface is a curve:*

$$\int_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{a} = \oint_C \vec{F} \cdot d\vec{l}.$$

To get the total swirl, I can work out the rotation at the boundary, or add up all the little whirlpools inside. Neighbouring whirlpools cancel along their shared edges, leaving only the boundary.

Theorem 3 (Gauss' divergence theorem). *The boundary of a volume is a surface:*

$$\int_V (\vec{\nabla} \cdot \vec{F}) d^3\vec{r} = \oint_S \vec{F} \cdot d\vec{a}.$$

To find the total flux leaking out through the surface, I can add up all the sources inside. Flux between neighbouring cells cancels, leaving only the outer surface.

Definition 14 (Current density). *The current I through a wire is not a local quantity, since a wire has finite thickness. The local version is the current density*

$$\vec{J} = \frac{d\vec{I}}{da_{\perp}},$$

where $da_{\perp}$ is the differential cross section perpendicular to the flow.

Remark 4. *The divergence theorem 3 applied to $\vec{J}$ is how the conservation of charge gets written down.*

2.5 Second Derivatives Identities

Definition 15 (Laplacian). *Taking the divergence of a gradient gives*

$$\vec{\nabla} \cdot (\vec{\nabla} f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \equiv \nabla^2 f,$$

where ∇^2 is the Laplacian¹.

¹Mathematicians often write Δ , but in physics we use ∇^2 .

Example (Sign of the Laplacian). *At a local maximum $\vec{\nabla}f$ points inward from every side, so it has negative divergence and $\nabla^2 f < 0$; at a minimum, $\nabla^2 f > 0$. At a saddle the second derivatives can cancel, so a flat point alone tells you nothing.*

The full list of second derivative identities:

Divergence of a gradient. $\vec{\nabla} \cdot (\vec{\nabla}f) = \nabla^2 f$.

Curl of a gradient. $\vec{\nabla} \times (\vec{\nabla}f) = 0$.

Divergence of a curl. $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$.

Identity 1. $\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$.

2.6 Index Notation

Remark 5. *A matrix carries two indices, A_{ij} , where i says which row and j says which column.*

Definition 16 (Transpose). *The transpose swaps the indices, $(A^T)_{ij} = A_{ji}$.*

Definition 17 (Symmetry). *A symmetric matrix has $A_{ij} = A_{ji}$, and an anti-symmetric one has $B_{ij} = -B_{ji}$.*

Definition 18 (Kronecker delta). *The Kronecker delta equals +1 when its condition is fulfilled and 0 otherwise, so it could be the identity matrix in index notation:*

$$\delta_{ij} = \begin{cases} +1 & i = j \\ 0 & i \neq j \end{cases}.$$

Example.

$$\sum_j \delta_{ij} v_j = v_i.$$

Proposition 1 (Matrix Multiplication).

$$\vec{w} = A\vec{v} \implies w_i = \sum_j A_{ij} v_j.$$

Proposition 2 (Dot Product).

$$\vec{v} \cdot \vec{w} = \sum_i v_i w_i.$$

Proposition 3 (Cross Product).

$$(\vec{v} \times \vec{w})_i = \sum_{j,k} \varepsilon_{ijk} v_j w_k.$$

Definition 19 (Levi-Civita symbol). ε_{ijk} is defined by

- $\varepsilon_{123} = +1$, and cyclic permutations keep the sign: $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312}$.
- Swapping any two indices flips the sign, so $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$.
- If any two indices are equal, $\varepsilon_{ijk} = 0$.

Example. For $i = 1$, only the terms with $(j, k) = (2, 3)$ and $(3, 2)$ survive:

$$(\vec{v} \times \vec{w})_1 = \varepsilon_{123}v_2w_3 + \varepsilon_{132}v_3w_2 = v_2w_3 - v_3w_2.$$

Proposition 4 (Epsilon-delta identity). Contracting two Levi-Civita symbols on a shared index gives

$$\sum_i \varepsilon_{ijk} \varepsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}.$$

2.7 Proving the Identities

2.7.1 Curl of a Curl

Set $\vec{G} \equiv \vec{\nabla} \times (\vec{\nabla} \times \vec{F})$. Then,

$$G_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \sum_{m,n} \varepsilon_{kmn} \partial_m F_n = \sum_{j,k,m,n} \varepsilon_{ijk} \varepsilon_{kmn} \partial_j \partial_m F_n.$$

The shared index k is in the third slot of the first symbol, so rotate it forward with the cyclic property $\varepsilon_{ijk} = \varepsilon_{kij}$ to match the proposition (k now first in both):

$$\sum_k \varepsilon_{kij} \varepsilon_{kmn} = \delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}.$$

Reading off the proposition with the summed index k removed: the leftover pairs are (i, j) and (m, n) , giving straight $(\delta_{im} \delta_{jn})$ minus crossed $(\delta_{in} \delta_{jm})$. Hence

$$G_i = \sum_{j,m,n} (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \partial_j \partial_m F_n.$$

Each Kronecker delta collapses one sum by setting its indices equal. In the first term δ_{im} forces $m = i$ and δ_{jn} forces $n = j$; in the second, δ_{in} forces $n = i$ and δ_{jm} forces $m = j$:

$$G_i = \sum_j \partial_j \partial_i F_j - \sum_j \partial_j \partial_j F_i.$$

The first term is $\partial_i \sum_j \partial_j F_j = \left(\vec{\nabla} (\vec{\nabla} \cdot \vec{F}) \right)_i$ and the second is $-(\nabla^2 \vec{F})_i$, giving $\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$.

2.7.2 Curl of a Gradient

Start from the definition, with $\vec{\nabla} f = \partial_k f$:

$$\left(\vec{\nabla} \times (\vec{\nabla} f) \right)_i = \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f.$$

Call this sum S . Since j and k are both summed over, they are dummy labels and can be renamed freely; swap $j \leftrightarrow k$ everywhere:

$$S = \sum_{j,k} \varepsilon_{ikj} \partial_k \partial_j f.$$

Now clean up the two factors. Antisymmetry of ε gives $\varepsilon_{ikj} = -\varepsilon_{ijk}$, and partial derivatives commute so $\partial_k \partial_j f = \partial_j \partial_k f$. Hence

$$S = - \sum_{j,k} \varepsilon_{ijk} \partial_j \partial_k f = -S.$$

So $S = -S$, which forces $S = 0$. Therefore $\vec{\nabla} \times (\vec{\nabla} f) = 0$.

2.8 Irrotational and Solenoidal Fields

The two vanishing identities let us classify fields.

- Since $\vec{\nabla} \times (\vec{\nabla} \psi) = 0$, writing $\vec{F} = \vec{\nabla} \psi$ guarantees $\vec{\nabla} \times \vec{F} = 0$. Such a field is **irrotational**.
- Since $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$, writing $\vec{F} = \vec{\nabla} \times \vec{A}$ guarantees $\vec{\nabla} \cdot \vec{F} = 0$. Such a field is **solenoidal**.

Any vector field can be written as $\vec{\nabla} \psi + \vec{\nabla} \times \vec{A}$.

Theorem 4 (Irrotational fields). *The following four statements are equivalent:*

- (i) $\vec{\nabla} \times \vec{F} = 0$ everywhere.
- (ii) $\int_a^b \vec{F} \cdot d\vec{l}$ is independent of path.
- (iii) $\oint_C \vec{F} \cdot d\vec{l} = 0$ for any closed loop.
- (iv) $\vec{F} = \vec{\nabla} \psi$ for some scalar ψ .

Theorem 5 (Solenoidal fields). *The following four statements are equivalent:*

- (i) $\vec{\nabla} \cdot \vec{F} = 0$ everywhere.
- (ii) $\int_S \vec{F} \cdot d\vec{a}$ is independent of the surface, for surfaces sharing a boundary.
- (iii) $\oint_S \vec{F} \cdot d\vec{a} = 0$ for any closed surface.

(iv) $\vec{F} = \vec{\nabla} \times \vec{A}$ for some vector field $\vec{A}$.

Remark 6. In electrostatics we will find $\vec{\nabla} \times \vec{E} = 0$. That is the irrotational case, and it is what licenses us to write $\vec{E} = -\vec{\nabla}V$ and introduce the electric potential. In magnetostatics we will find $\vec{\nabla} \cdot \vec{B} = 0$, which is the solenoidal case and licenses $\vec{B} = \vec{\nabla} \times \vec{A}$, the vector potential.

3 Electrostatics

3.1 Gauss' Law and the Inverse Square Law

Coulomb's law for two point charges,

$$F = k \frac{q_1 q_2}{r^2},$$

has the same $1/r^2$ shape as Newton's gravity, $F = Gm_1 m_2 / d^2$. The reason is geometric rather than a coincidence, and Gauss' law makes that explicit.

Law 1 (Gauss' law). *The flux of $\vec{E}$ through any closed surface is set by the charge inside it:*

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}.$$

Surround a point charge with a sphere of radius r . The flux is fixed by Q , but the surface area over which it is spread grows like r^2 , so $E \propto 1/r^2$. The exponent is really a count of dimensions.

Example (Coulomb's law in d dimensions). *In a universe with four spatial dimensions (x, y, z, w) , volumes are four-dimensional and the surfaces enclosing them are three-dimensional, with "area" scaling like r^3 . Gauss' law then gives $E \propto 1/r^3$. In general, in d spatial dimensions a point charge has $E \propto 1/r^{d-1}$. The inverse square law is what three dimensions look like.*

Remark 7 (Compactified dimensions). *Suppose an extra dimension exists but is curled up, the way a flat sheet with coordinates (x, y) can be rolled into a cylinder whose second coordinate w is periodic with a small radius. At distances much smaller than that radius, field lines spread out into all the dimensions and the field falls off faster than $1/r^2$. At distances much larger than it, there is only room to spread out in the large dimensions and the usual $1/r^2$ is recovered. So a measured deviation from the inverse square law at short range would be evidence for extra dimensions.*

This has been tested for gravity by the Eöt-Wash torsion balance experiments (2004, 2007, 2020), which checked the $1/r^2$ law down to separations of $52 \mu\text{m}$. Newtonian gravity fit the data, ruling out gravitational strength Yukawa interactions with ranges above $38.6 \mu\text{m}$ at 95% confidence.

Remark 8 (Field lines). *Field lines start on positive charges and end on negative ones, and their density measures the field strength. This is the same statement as Gauss' law: the number of lines through a closed surface counts the*

enclosed charge. Two like charges have lines that bend away from each other. An infinite sheet of charge has lines that cannot spread out at all, so its field is uniform and does not fall off with distance.

3.2 Gauss' Law with Symmetry

Gauss' law comes in two equivalent forms, related by the divergence theorem:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0} \iff \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0},$$

where ρ is the charge density. Both are always true, but the integral form is only useful for computing $\vec{E}$ when the symmetry of the charge distribution lets us pull E out of the integral. That requires a closed surface on which $\vec{E}$ has constant magnitude and is either parallel or perpendicular to $d\vec{a}$ everywhere.

Remark 9. The integral form answers the question "what is the flux through some closed surface?" for any surface at all, whatever the field looks like: only the enclosed charge matters. It is turning that flux back into a value of $\vec{E}$ at a point that needs symmetry.

3.2.1 Spherical symmetry

Example (Point charge). Surround a point charge q with a sphere S of radius r centred on it. Two facts reduce the integral to a product:

- By symmetry, $\vec{E} = E(r)\hat{r}$: the field is radial and has the same magnitude everywhere on S .
- By geometry, $d\vec{a} = \hat{n} da = \hat{r} da$ on a sphere.

So $\vec{E} \cdot d\vec{a} = E da$, and

$$\oint_S \vec{E} \cdot d\vec{a} = E \underbrace{\oint_S da}_{4\pi r^2} = \frac{q}{\epsilon_0} \implies E = \frac{q}{4\pi\epsilon_0 r^2}.$$

This is Coulomb's law, with $k = 1/4\pi\epsilon_0$.

3.2.2 Cylindrical symmetry

Example (Infinite line of charge). Take an infinite straight line carrying charge λ per unit length. Use as Gaussian surface a cylinder of radius z and length d coaxial with the line. By symmetry $\vec{E}$ points radially outward from the line, perpendicular to the curved surface, and it has no component along the line. It is therefore parallel to $d\vec{a}$ on the curved surface and perpendicular to $d\vec{a}$ on the two end caps, so the end caps contribute no flux. The enclosed charge is λd , and

$$\oint_S \vec{E} \cdot d\vec{a} = E \underbrace{\int_{\text{curved}} da}_{2\pi z d} = \frac{\lambda d}{\epsilon_0} \implies E = \frac{\lambda}{2\pi\epsilon_0 z}.$$

The length d drops out, as it must for an infinite line. The field falls off as $1/z$, not $1/z^2$.

Example (Finite versus infinite line). Compare a line of length $2L$ with an infinite line, both with the same λ , and ask which gives the stronger field at a point P just above the midpoint. The infinite line does. Draw the same cylinder around the middle of each: $Q_{enc} = \lambda d$ is the same, so the total flux is the same. For the infinite line no flux leaves through the end caps by symmetry. For the finite line the field lines near the ends bend outward, so some flux "leaks" out through the end caps. With the same total, the flux through the curved surface, and hence E at P , must be smaller for the finite line.

3.2.3 Why the fall-off depends on the geometry

The exponent in the fall-off counts how much charge a fixed cone of field lines can "see" as it grows. Picture a narrow cone opening out from the observation point at distance r toward the source. The cone's cross section grows like r^2 , which is the $1/r^2$ dilution of a point charge. But for extended sources the enclosed charge grows too:

- A line of charge: the cone intercepts a length of line proportional to r , so the field goes like $r/r^2 = 1/r$.
- A sheet of charge: the cone intercepts an area proportional to r^2 , so the field goes like r^2/r^2 , a constant. This is the uniform field of the infinite sheet noted above.

Remark 10. For two equal positive charges, the field lines from each are pushed away from the other and none cross the midplane between them. On that plane the field is purely parallel to the plane, and it vanishes at the midpoint. A Gaussian surface straddling the midplane still obeys Gauss' law, but the field on it is not uniform, so the law cannot be used to solve for $\vec{E}$ there: no symmetry, no shortcut.

3.3 Coulomb's Law for Continuous Distributions

Superpose the point charge field over the source:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}') d^3\vec{r}'}{\vec{r}^2} \hat{\mathbf{t}}, \quad \vec{r} \equiv \vec{r} - \vec{r}',$$

where $\vec{r}$ is the field point, $\vec{r}'$ the source point, and $\vec{r}$ (Griffiths' script r) the separation vector from source to field point. For a line of charge the volume element collapses to $\rho d^3\vec{r}' \rightarrow \lambda(\vec{r}') dl'$.

Example (Finite line of charge by direct integration). A line of length $2L$ lies along the x axis from $-L$ to L with uniform λ , and we want $\vec{E}$ at $P = (0, 0, z)$ on the perpendicular bisector.

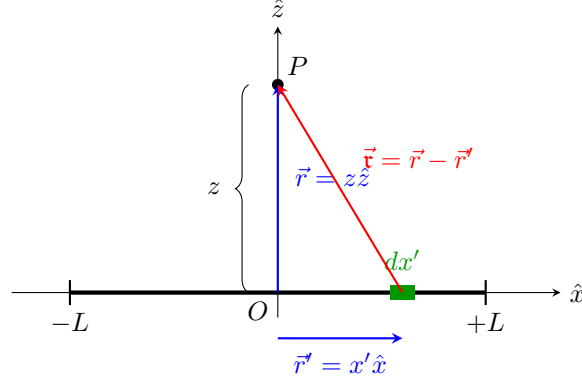


Figure 1: Setup for the finite line of charge. The source point $\vec{r}' = x'\hat{x}$ runs from $-L$ to L ; the field point is P at $\vec{r} = z\hat{z}$; the separation vector is $\vec{r} = \vec{r} - \vec{r}'$.

Here $\vec{r} = z\hat{z}$ and $\vec{r}' = x'\hat{x}$, so

$$\vec{r} = z\hat{z} - x'\hat{x}, \quad |\vec{r}| = \sqrt{z^2 + x'^2}, \quad \hat{t} = \frac{z\hat{z} - x'\hat{x}}{\sqrt{z^2 + x'^2}},$$

and the total charge is $Q = \int \lambda dx' = 2\lambda L$. Feeding this into the integral,

$$\vec{E}(z\hat{z}) = \frac{\lambda}{4\pi\epsilon_0} \int_{-L}^L \frac{1}{z^2 + x'^2} \frac{z\hat{z} - x'\hat{x}}{\sqrt{z^2 + x'^2}} dx'.$$

The $\hat{x}$ term is odd in x' and integrates to zero, as the symmetry of the setup demands. For the $\hat{z}$ term, $\int dx'/(z^2 + x'^2)^{3/2} = x'/(z^2\sqrt{z^2 + x'^2})$, so

$$\vec{E} = \frac{\lambda z\hat{z}}{4\pi\epsilon_0} \cdot \frac{2L}{z^2\sqrt{L^2 + z^2}} = \underbrace{\frac{\lambda}{2\pi\epsilon_0 z}}_{\text{infinite line}} \cdot \underbrace{\frac{L}{\sqrt{L^2 + z^2}}}_{< 1}.$$

The prefactor is the infinite line result from Gauss' law, and the second factor is the correction for finite length. It is always less than one, which is the “leaked flux” argument made quantitative. Two limits:

- $z \ll L$: $L/\sqrt{L^2 + z^2} \approx L/L = 1$, so $\vec{E} \rightarrow \frac{\lambda}{2\pi\epsilon_0 z} \hat{z}$. Close to the line it looks infinite, and Gauss' law is recovered.
- $z \gg L$: $\sqrt{L^2 + z^2} \approx z$, so $\vec{E} \rightarrow \frac{2\lambda L}{4\pi\epsilon_0 z^2} \hat{z} = \frac{Q}{4\pi\epsilon_0 z^2} \hat{z}$. Far away it looks like a point charge $Q = 2\lambda L$.

Remark 11. Always check limits. A result for a finite distribution should collapse to a simpler known field both up close and far away; if it does not, the integral is wrong.

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